CULTIVATION GUIDE | FIELD MONITORING

Reading Satellite Vegetation Data for Paddy Fields

What each value from the satellite service means, how to classify it, and the caveats that apply to Sri Lankan paddy.

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Teaching edition for the STEMLink AI Engineer Bootcamp. Content restructured from AgriVibe knowledge base AV-KB-001 Rev 1.1, which grounds its agronomy in Department of Agriculture and RRDI guidance for Sri Lanka. Not for use with real farmers until the verification items in AV-POL-03 are closed.

THRESHOLD CONVENTION

Numeric thresholds in this guide are project demonstration thresholds defined for AgriVibe. They are not Department of Agriculture recommendations and not universal agronomic thresholds.

1. Current mean NDVI

The Normalized Difference Vegetation Index (NDVI) ranges between -1 and +1 and measures vegetation greenness from the difference between near-infrared and red-light reflectance. Higher positive values generally indicate more active or dense green vegetation. Lower values may indicate sparse vegetation, bare ground, water or stressed vegetation. The current_mean_ndvi value is the average greenness across the selected field for the current observation period.

NDVI must not be treated as a disease diagnosis or as a measurement of yield. Change over time carries far more weight than the absolute value. A mean NDVI of 0.58 is neither good nor bad on its own. It acquires meaning only against the same field's previous value, the crop's growth stage, weather conditions, data quality and the variability across the field.

STANDING WATER CAVEAT

Lowland rice in Sri Lanka is normally grown with standing water in the field. Water reflects very differently from vegetation, so a field holding more water than usual can show a lower mean NDVI even when the plants themselves are unchanged. A drop in NDVI must never be read as proof that plants have deteriorated when the field may simply be wetter.

2. NDVI change percentage

The satellite service computes the change between the current and previous observation periods as: (current mean NDVI - previous mean NDVI) / previous mean NDVI x 100.

Worked example: previous mean NDVI 0.66, current mean NDVI 0.56. Change = (0.56 - 0.66) / 0.66 x 100 = -15.2 percent, which is a significant decline.

Table 1. Change classification (project demonstration thresholds)

Below -15%. Classification: Significant decline; Trend label: significant_decline.
-15% to -5%. Classification: Moderate decline; Trend label: moderate_decline.
-5% to +5%. Classification: Relatively stable; Trend label: stable.
+5% to +15%. Classification: Moderate increase; Trend label: moderate_increase.
Above +15%. Classification: Significant increase; Trend label: significant_increase.

A negative change means average greenness has declined relative to the previous period; a positive change means it has increased. The change says nothing about cause. A decline may result from crop stress, harvesting, natural maturity, waterlogging, drought, disease, pest damage, land disturbance, increased standing water, or measurement limitations.

3. Growth stage and NDVI

NDVI in a healthy rice crop rises through tillering, peaks around panicle initiation and booting, then falls naturally as the crop ripens toward harvest. A decline in a crop older than roughly 75 to 80 days may be normal ripening rather than a problem. Crop age must be checked before a decline is treated as a warning sign.

Table 2. Approximate crop age to growth stage, with expected NDVI

0 to 2 weeks. Stage: Establishment; NDVI expectation: Low; much water and soil visible.
2 to 6 weeks. Stage: Tillering; NDVI expectation: Rising.
6 to 9 weeks. Stage: Panicle initiation to booting; NDVI expectation: Near peak.
9 to 12 weeks. Stage: Flowering to grain filling; NDVI expectation: High, then beginning to fall.
Final weeks. Stage: Ripening; NDVI expectation: Falling naturally toward harvest.

4. Trend label

The trend field is a human-readable label derived from the change percentage, taking one of five values: significant_decline, moderate_decline, stable, moderate_increase or significant_increase. The numerical change remains the primary evidence; the label is a convenience for presentation.

5. Field uniformity

The service returns the NDVI standard deviation across the field and converts it to a uniformity label using the coefficient of variation (CV), calculated as standard deviation divided by mean NDVI.

Table 3. Uniformity classification (project demonstration thresholds)

Up to 0.10: uniform
0.10 to 0.20: moderately_variable
Above 0.20: uneven

A uniform result means values are relatively similar throughout the field; an uneven result means there is considerable variation between parts of it. Uneven vegetation does not reveal a cause and does not mean the whole field is unhealthy.

In Sri Lankan paddy fields, uneven patterns commonly follow the lie of the land: low-lying corners that retain water, sections near the bund, areas where interflow enters from adjacent highland, and parts with poor surface drainage. Recommending inspection of the weaker-looking sections specifically is more useful to the farmer than a whole-field statement.

6. Valid observations

This is the count of usable satellite observation dates remaining after cloud and quality filtering.

Table 4. Evidential weight by observation count

4 or more: Stronger time-series evidence
2 to 3: Limited but potentially usable
Fewer than 2: Insufficient for a reliable trend

Cloud cover during the North-East monsoon (roughly September to March, the Maha season) and the South-West monsoon (roughly May to August, the Yala season) frequently removes usable observations. A low count during these periods is expected and is not itself evidence that anything is wrong with the field.

7. Data quality

The data_quality field takes the values good, moderate or poor. It is derived from the number of valid observations, cloud coverage, the share of the field with usable pixels, and whether the current and previous periods are comparable.

MANDATORY RULE

If data quality is poor, or valid observations number fewer than two, no strong vegetation conclusion may be offered. State that the available satellite evidence is insufficient and recommend waiting for a further observation or carrying out a field inspection. An honest statement that the evidence is not strong enough is always preferred to a confident answer built on weak data.